

Closed-graph continuity $\Rightarrow$ Mixture continuity

Result — ai; produced by Claude Fable 5.1 (Cowork session).

0.1 Premises

- **Closed-graph continuity.** The set $\{(p, q) \in \Delta(X) \times \Delta(X) : p \succcurlyeq q\}$ is closed in the product topology.

0.2 Conclusion

- **Mixture continuity.** For all $p, q, r \in \Delta(X)$ the sets $\{\lambda \in [0,1] : \lambda p + (1-\lambda)q \succcurlyeq r\}$ and $\{\lambda \in [0,1] : r \succcurlyeq \lambda p + (1-\lambda)q\}$ are closed.

0.3 Proof

$\lambda \mapsto (\lambda p + (1-\lambda)q, r)$ and $\lambda \mapsto (r, \lambda p + (1-\lambda)q)$ are continuous maps $[0,1] \rightarrow \Delta(X) \times \Delta(X)$; the two sections are their preimages of the closed graph of $\succcurlyeq$.

Record: topics/decision-theory/results/closed-graph-implies-mixture-continuity.yaml